MAHATMA GANDHI UNIVERSITY ,KOTTAYAM

MGU-UGP (HOMOURS)REGULAR EXAMINATIONS MARCH 2026

FOURTH SEMESTER

DISCIPLINE SPECIFIC COURSE(DSC)-MG4DSCECT204-COMPUTER ORGANIZATION

2024 ADMISSION ONWARDS

DURATION :1 HOUR 30 MINUTES MAXIMUM MARKS:50

PART A

VERY SHORT ANSWER QUESTIONS

EACH QUESTION CARRIES 1 MARKS

1. Any statement related to EEPROM or Flash Memory.
2. (11)H
3. Full adder truth table.
4. A subtractor unit is a **combinational logic circuit** designed to perform the arithmetic subtraction of two binary numbers.
5. Reduce CPU involvement in I/O operations.
6. Any term related with cell quantity and layout.
7. A data bus is a bidirectional system pathway that transfers actual data between the CPU, memory (RAM), and input/output (I/O) components, allowing them to communicate.
8. Truth table of xnor gate.
9. 2
10. Toggle condition.

PART B

SHORT ANSWER QUESTIONS

EACH QUESTION CARRIES 5 MARKS

11. Advantages of Semiconductor RAM:

    High Speed: Access times are very fast, often measured in nanoseconds.

    Efficient Performance: Ideal for CPU cache (SRAM) and main memory (DRAM) due to low latency.

    Compact & Lightweight: Small size makes them ideal for smartphones, laptops, and tablets.

Low Power Consumption: Generally consume less power compared to other storage types.

Reliability: No moving parts mean they are more durable and shockproof.

Disadvantages of Semiconductor RAM:

Volatility: Data is lost when the power is turned off.

High Cost: Typically more expensive per gigabyte compared to non-semiconductor storage technologies.

Lower Capacity: While improving, they have lower storage densities compared to Hard Disk Drives (HDDs).

Limited Lifespan (DRAM): Requires constant power and periodic refresh cycles (for DRAM) to maintain data.

Sensitivity: Semiconductor chips are sensitive to environmental factors like high temperature and radiation.

12. Binary subtraction using 2's complement is performed by adding the minuend to the 2's complement of the subtrahend.

    The key steps are: 1. Find the 2's complement of the subtrahend (invert bits and add 1).

    2. Add this to the minuend.

    3. If a carry occurs, discard it; the result is positive.

    4. If no carry occurs, the result is negative and in 2's complement form.

    NOTE: IF A STUDENT EXPLAINED IT ONLY WITH EXAMPLES, AWARD MARKS IF THE STEPS ARE CORRECT.

13. Give full marks if the waveform is correct.
14. Keywords to check; - placement of data , complexity , speed, size.

    Any terms related with mapping techniques.

15. BCD definition with example– 2 marks

    BCD addition rule. Invalid BCD numbers.- 3 marks.

16. Definition – 2 marks. Explanation -3 marks.

    There are two primary types of parity generators:

Even Parity Generator: It produces a parity bit such that the total number of 1s in the code word is even.

If the data has an odd number of 1s, the generator sets the parity bit to 1.

If the data has an even number of 1s, the generator sets the parity bit to 0.

Odd Parity Generator: It produces a parity bit such that the total number of 1s in the code word is odd.

If the data has an even number of 1s, the generator sets the parity bit to 1.

If the data has an odd number of 1s, the generator sets the parity bit to 0.

PART C

ESSAY QUESTIONS

EACH QUESTION CARRIES 10 MARKS

17. Different number systems and integer representation with examples.

18. A) A+ABA+ABA+AB - 5 MARKS (WITH PROPER STEPS)

= A+B+B+AB (WHER A.A=A)

= A+B+AB (WHERE B+B=B)

= A(1+B)+B

=A+B (WHERE 1+B=1)

=A+B

B) A(A+B)A(A+B)A(A+B) - 5 MARKS (WITH PROPER STEPS)

=A.A.A.(A+B).(A+B).(A+B)

=A(A+B)

=A.A+AB

=A+AB

=A(1+B)

=A

19. A) (ABC)'+(D'+E)' -5 MARKS WITH PROPER STEPS

= A'+B'+C'+(D''.E')

=A'+B'+C'+DE'

B) (A+B+C)'+(D'E)' - 5 MARKS WITH PROPER STEPS.

= A'.B'.C'+(D''+E')

= A'B'C'+D+E'.